



Euclid x Rubin for Resolved Stellar Populations

Kayleigh Excell, UW Madison
DKM Stellar Streams Meeting
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Broader Context

Objective is to develop refined stellar samples for near-field cosmology using combination of Rubin (multiband photometry) + Euclid (space-based resolution)

Data Release Overlap Area

Rubin DP1 x Euclid Q1	→ $\sim 1 \text{ deg}^2$
Rubin DP2 (Jul-Sep 2026) x Euclid Q1	→ $\sim 30 \text{ deg}^2$
DES Y6 x Euclid DR1 (Oct 2026)	→ $\sim 1400 \text{ deg}^2$
Rubin DP2 x Euclid DR1	→ $\sim 30 \text{ deg}^2$ (DP2 wide concentrated along ecliptic)
LSST DR1 (mid 2028) x Euclid DR1	→ $\sim 1400 \text{ deg}^2$
LSST DR1 x Euclid DR2 (Mar 2029)	→ $\sim 5000 \text{ deg}^2$ (rough estimate)
LSST DR4 (mid 2031) x Euclid DR3 (Oct 2031)	→ $\sim 8000 \text{ deg}^2$

Broader Context



A few possible applications of Euclid with Rubin

Main topics for this presentation

(1) Use Euclid as “truth” catalog to evaluate Rubin-only stellar sample

→ requires accurate labels from Euclid; only valid where Euclid labels are reliable (probably Rubin $i < 24$)

(2) Use Euclid jointly with Rubin to develop refined Euclid+Rubin stellar sample

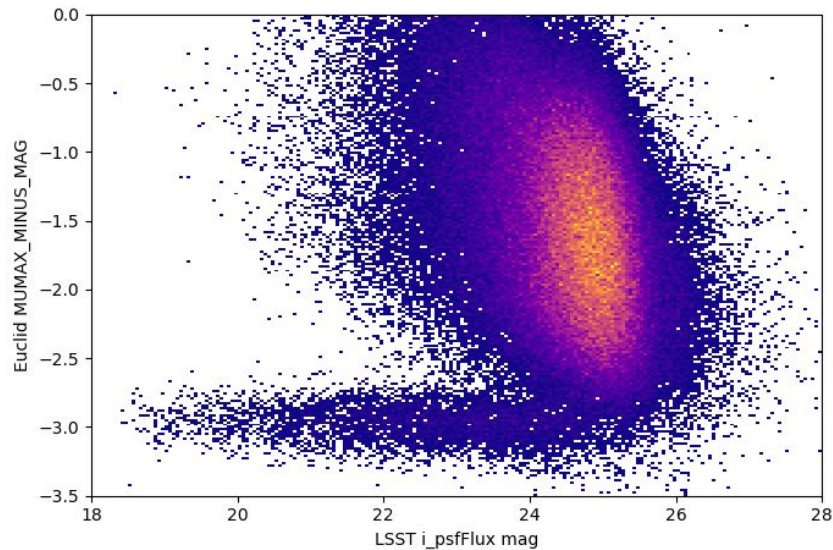
→ $\sim 30 \text{ deg}^2$ of overlap between Euclid Q1 and DP2 is sufficient for pipeline development; overlap area will grow to thousands of square degrees with subsequent data releases

(3) Use Euclid DR1 + DES Y6 as “pilot” for Euclid DR1 + LSST DR1 joint analysis (?)

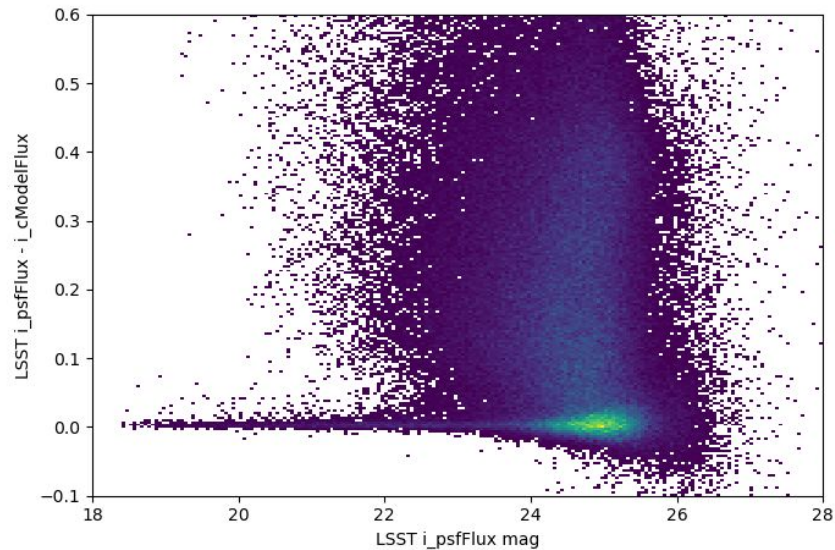
→ tests wide-area analysis challenges; possible discovery potential

All Objects in ECDFS Euclid vs Rubin Morphology

ECDFS Euclid Star-Galaxy Separation



ECDFS Rubin Star-Galaxy Separation



Euclid as Truth Catalog for Evaluating Rubin-Only Stellar Sample

Euclid's Star-Galaxy Classifier

POINT_LIKE_PROB

Value between 0 to 1

Only available for subset of sources (non-NIR)

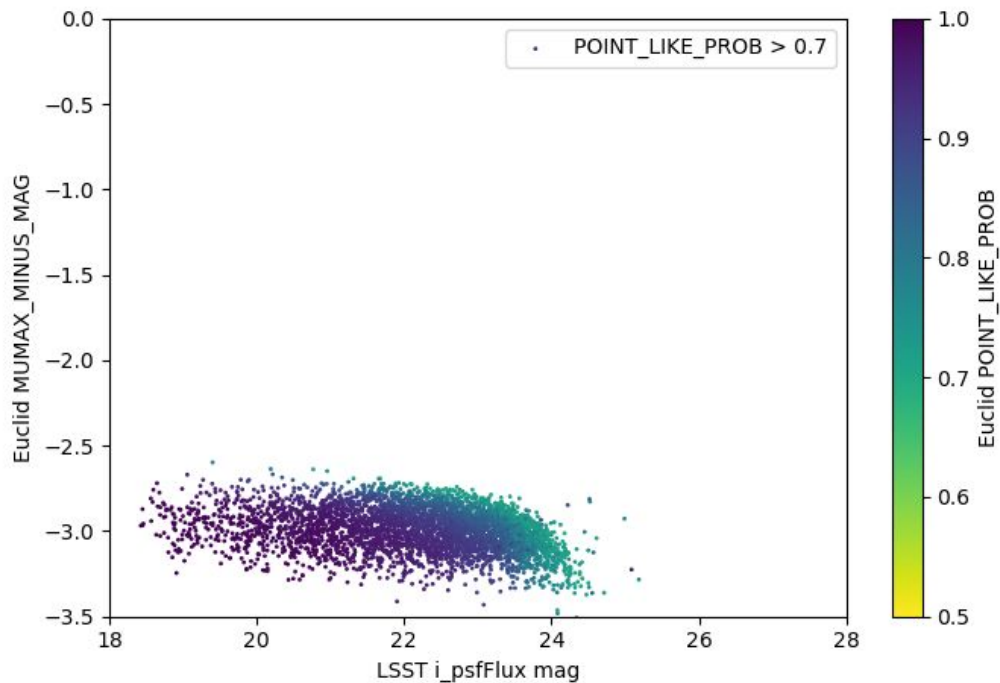
The Euclid catalog prioritized *purity* in their point-like flags/measurements
(according to [Žerjal et al. 2025](#))

Q1 papers, e.g. [Romelli et al. 2025](#), tend to use `POINT_LIKE_PROB > 0.7` to
classify point-like vs extended sources

This value is “inspired from the `SPREAD_MODEL` method provided by
SExtractor” but is calculated with `MUMAX_MINUS_MAG`

Note: `POINT_LIKE_FLAG`: Flag set to 1 for VIS detections with `DET_QUALITY_FLAG==0 AND POINT_LIKE_PROB>0.96`, otherwise set to NaN

POINT_LIKE_PROB cut

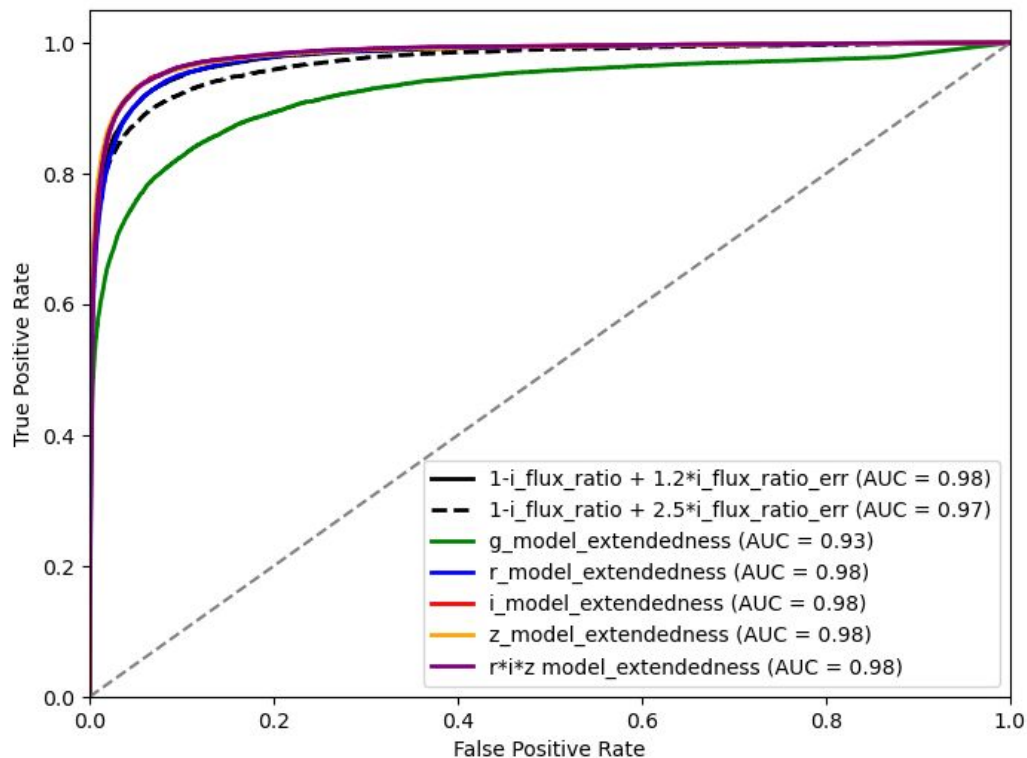


Data is from ECDFS

Valid down to $i \sim 24$ mag

→ Using this as truth we'll evaluate the performance of our Rubin classifiers, cutting the catalog at $i < 24$ mag

ROC Curves



Using matched area across ECDFS
and EDFS

Truth $\rightarrow$ POINT_LIKE_PROB > 0.7
Limited to i mag < 24

All classifiers seem to be converging
on the same performance

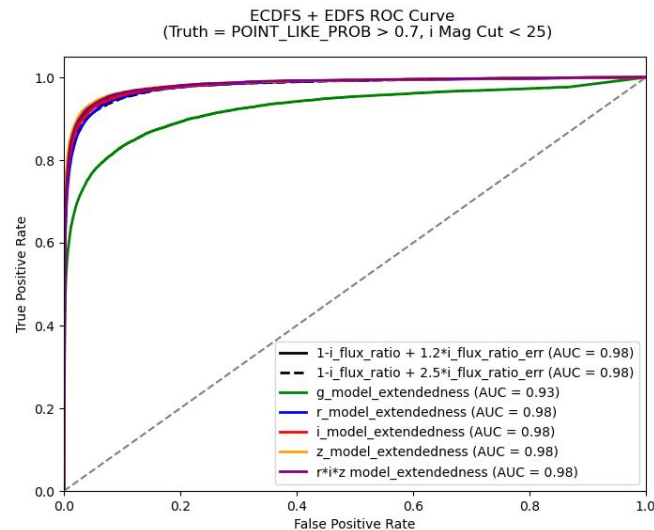
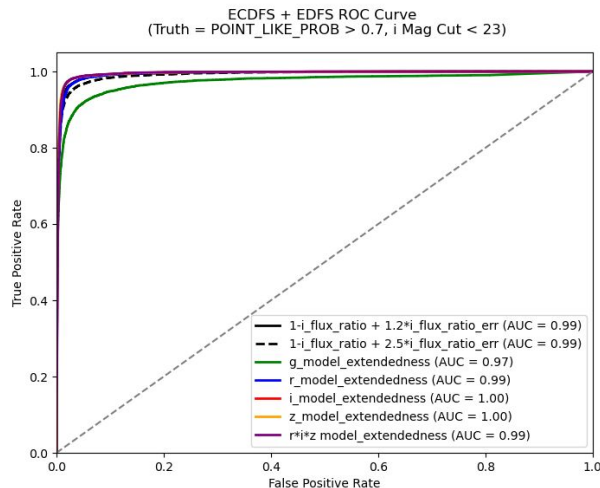
Except g

ROC Curves

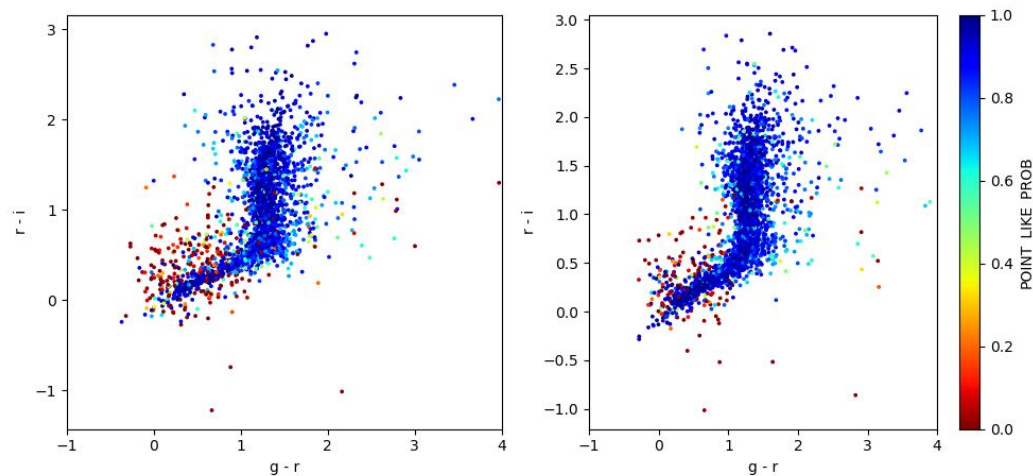
Classifiers seem to be doing well down to 23 mag

24/25 is where the performance seems to drop

Caveat: this Euclid truth label may be too pure so as to be not useful in evaluating Rubin classifier performance (see extra slides)

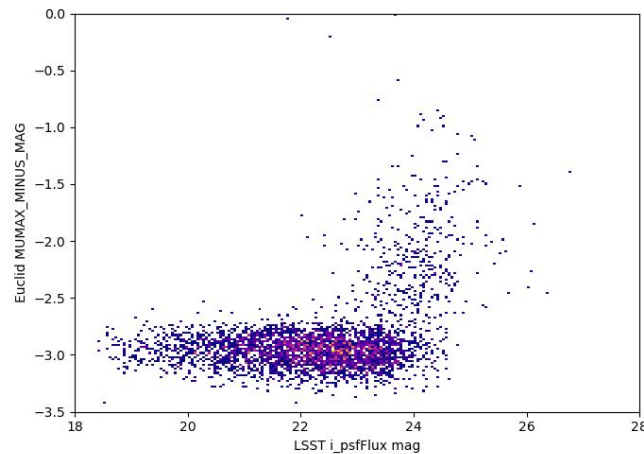


ECDFS vs EDFS $i_model_extendedness < 0.15$



Different thresholds in extra slides

It is possible to get a fairly pure Rubin-only stellar sample, but you must be rather restrictive to do it
→ only reaches down to ~ 24 mag, with contamination



Euclid Morphology and NIR + Rubin Photometry Stellar Sample

Euclid Morphology Cut

For a stellar population search, it may be decided that Euclid's `POINT_LIKE_PROB` is not complete enough

Another option suggested by [Žerjal et al. 2025](#) - photometric search for ultracool dwarfs

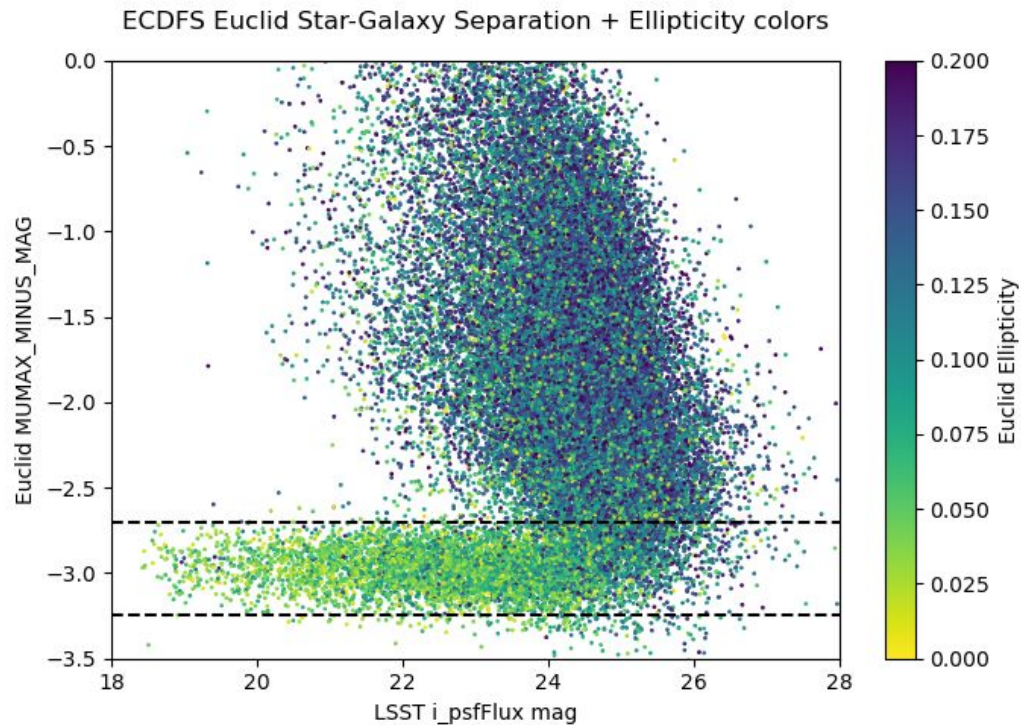
`(ELLIPTICITY < 0.2) & (-3.25 < MUMAX_MINUS_MAG < -2.7)`

`ELLIPTICITY` $\equiv 1-B/A$ $\leftarrow$ B is semi-minor axis, A is semi-major axis

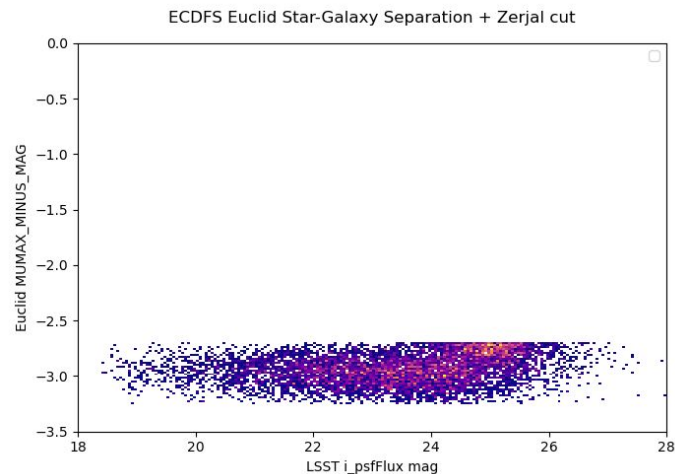
`MUMAX_MINUS_MAG` = compares the peak surface brightness mag. and total mag.,
as a measurement of concentration of light

ELLIPTICITY<0.2

& MUMAX_MINUS_MAG between -3.25 and -2.7

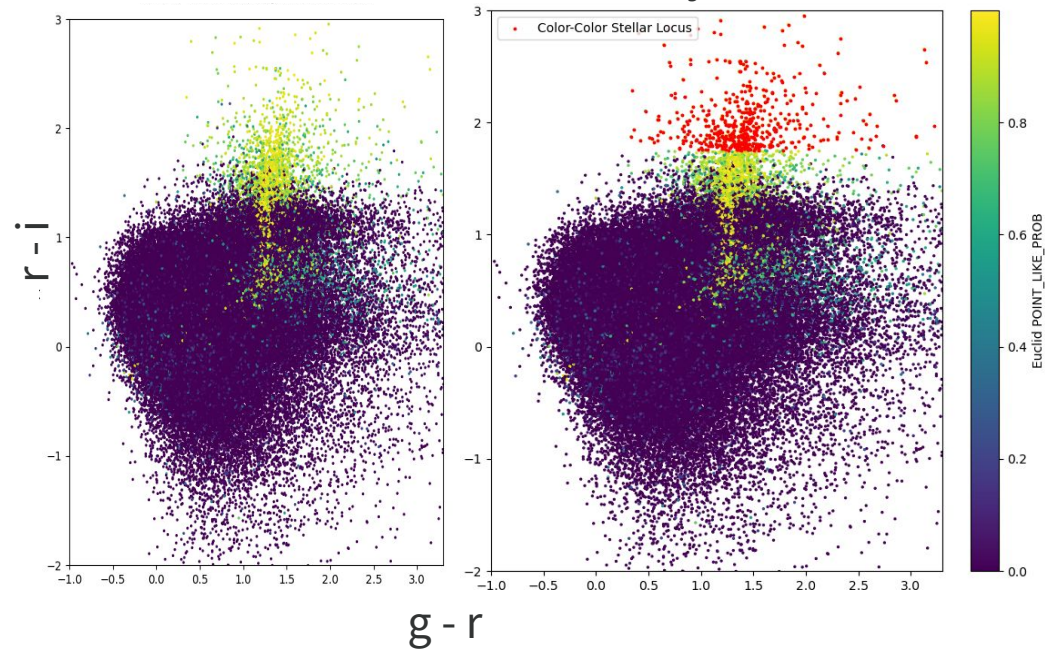


More complete
Allows down to ~26 mag

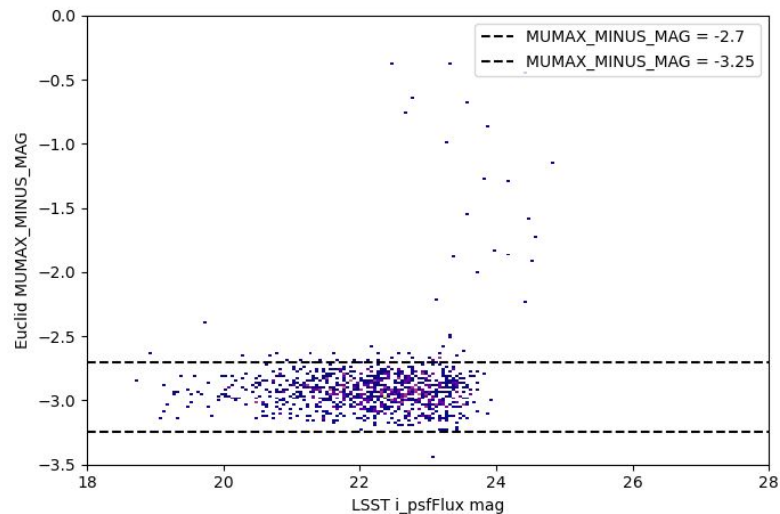


Stellar Locus Finding -> Morphology

EDFS & ECDFS All Objects

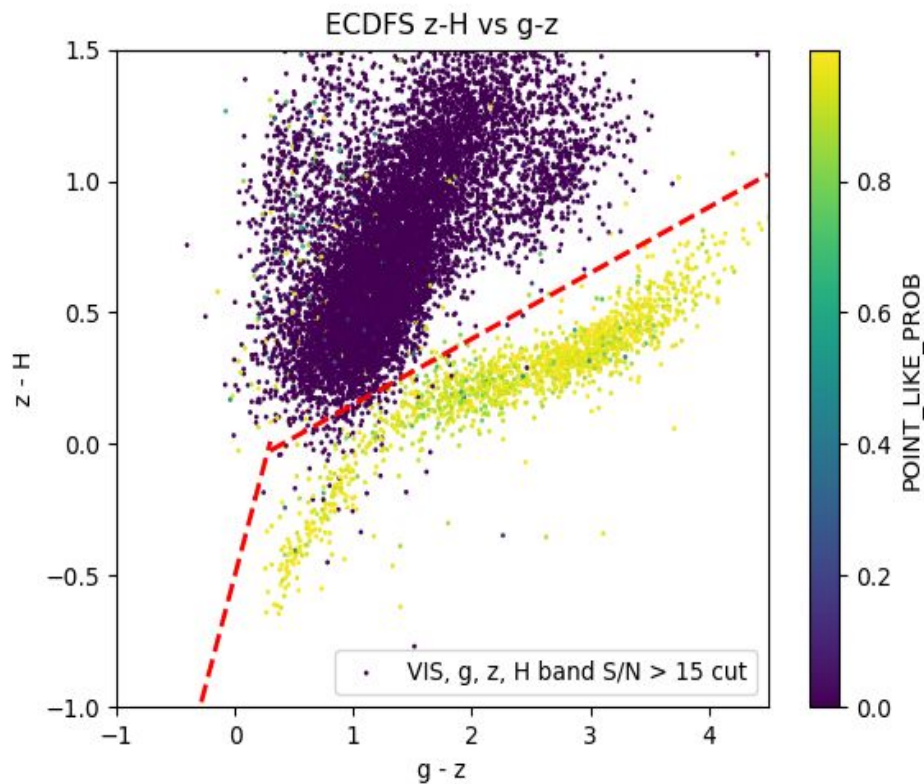


EDFS + ECDFS Stellar Locus Morphology Test



A simple $r-i$ mag > 1.75 cut
Agrees well with the Zerjal et al. cut

NIR + Optical Photometry



Infrared and optical color-color space might give opportunity for deeper separation

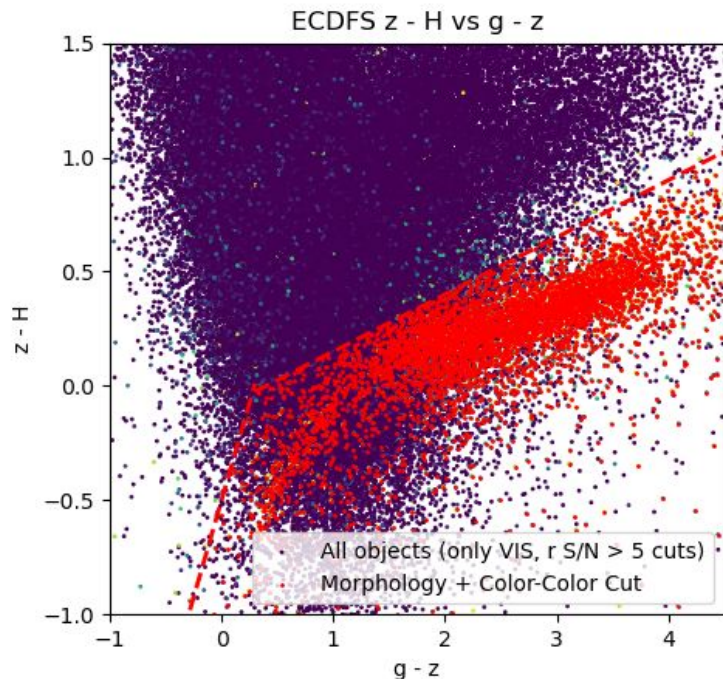
This particular cut on higher significance data (source-to-noise ratio > 15 in VIS, g, z, and H bands) allows for separation down to 23 mag

Cut chosen visually:

$(z-H) < -0.5 + 1.7 \cdot (g-z)$ for $-1 < g-z \leq 0.3$

$(z-H) < -0.1 + 0.25 \cdot (g-z)$ for $0.3 \leq g-z < 4.5$

Morphology + Color



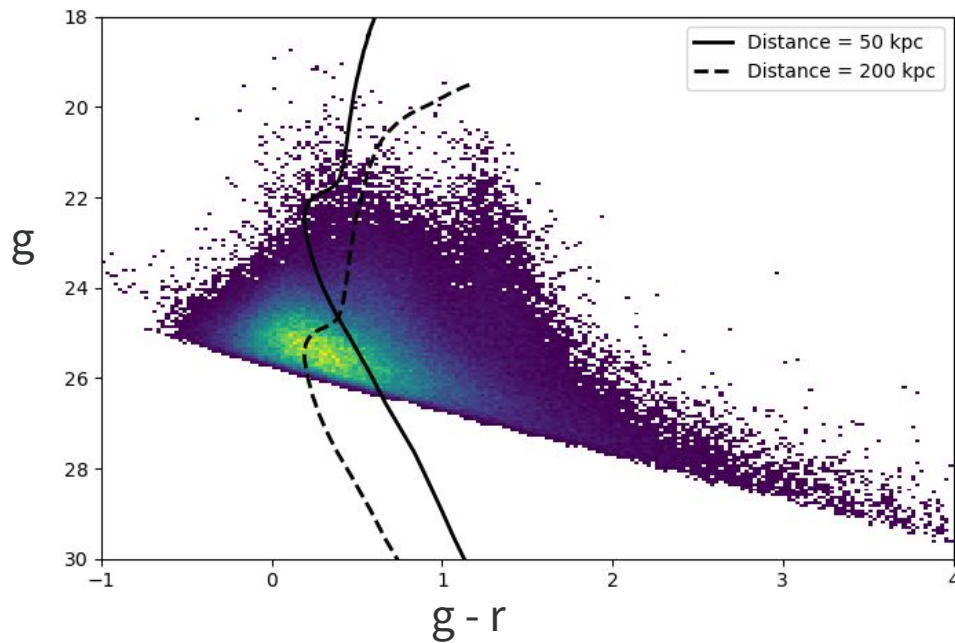
We then combine this color-color cut with a morphology cut based on ellipticity and MUMAX_MINUS_MAG

For all objects (only our normal S/N quality cuts in the VIS and r bands)

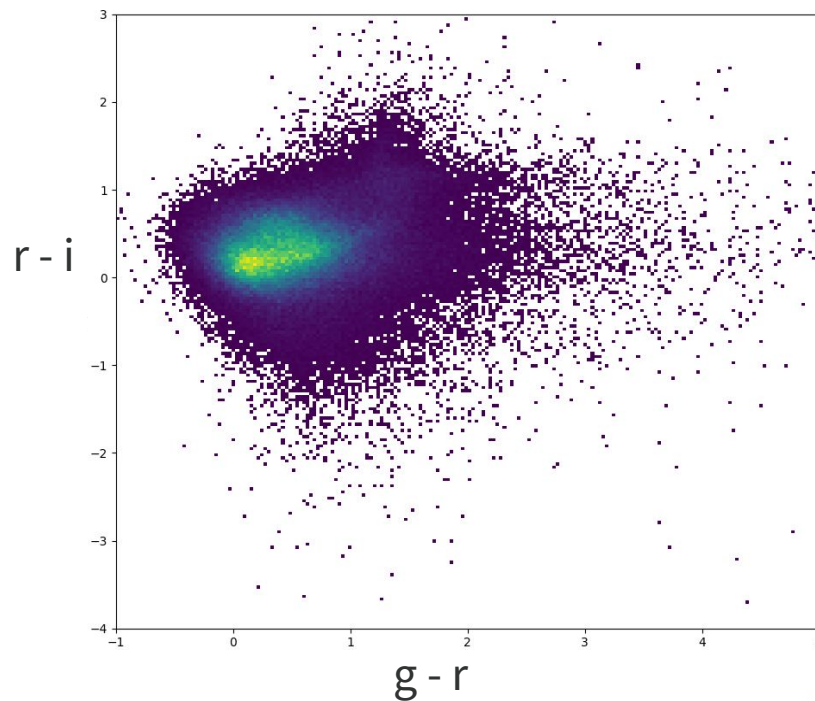
This extends separation down to $i \sim 25$ mag

ECDFS DP2 all objects

color-magnitude

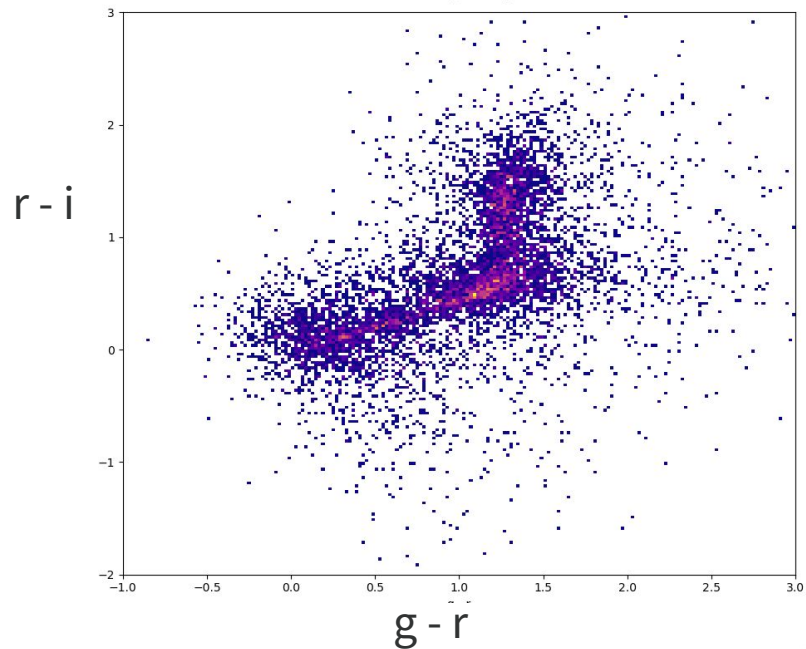
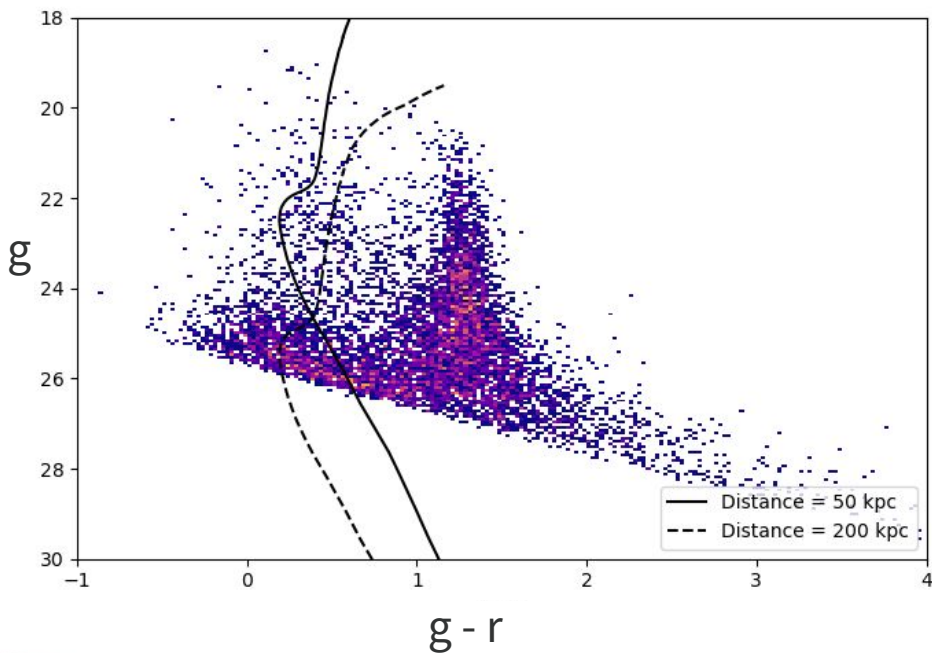


color-color

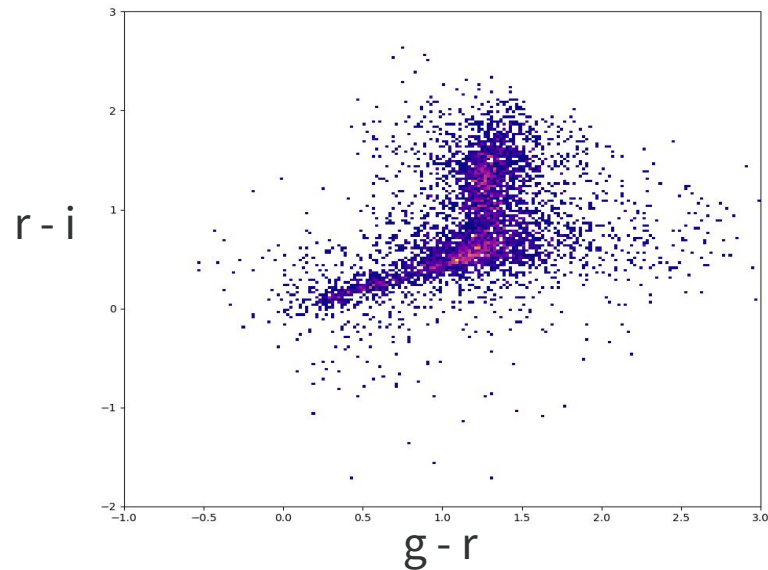
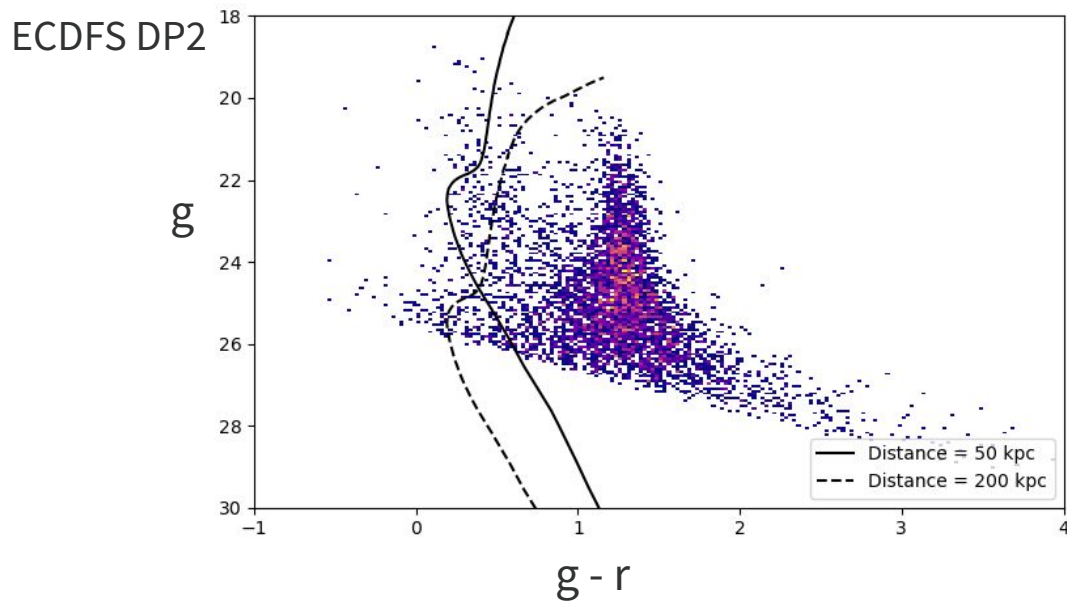


Euclid Morphology Cut

Selecting with ($\text{ELLIPTICITY} < 0.2$) & ($-3.25 < \text{MUMAX_MINUS_MAG} < -2.7$)



Euclid Morphology + Euclid x Rubin Color Cut



Cut summary:

$$(z-H) < -0.5 + 1.7 \cdot (g-z) \text{ for } -1 < g-z \leq 0.3$$

$$(z-H) < -0.1 + 0.25 \cdot (g-z) \text{ for } 0.3 \leq g-z < 4.5$$

$$(\text{ELLIPTICITY} < 0.2) \ \& \ (-3.25 < \text{MUMAX_MINUS_MAG} < -2.7)$$



Extra Slides

Euclid documentation

Catalog

https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/dpcards/mer_finalcatalog.html#merfinalcatalog

<https://euclid.esac.esa.int/dr/q1/dpdd/merdpd/mermorphologycookbook.html>

On stellar selection

<https://arxiv.org/pdf/2503.22497>

<https://arxiv.org/pdf/2503.15305> 7.4

extra slide quality cuts applied to data

LSST:

`detect_isIsolated == True`

`r_psfFlux/r_psfFluxErr > 5`

`For all bands {band}_psfFlux_flag == 0`

Euclid:

`FLUX_VIS_PSF/FLUXERR_VIS_PSF > 5`

`SPURIOUS_FLAG == 0`

`DET_QUALITY_FLAG == 0 or 2`

`For all bands FLAG_{band} == 0`

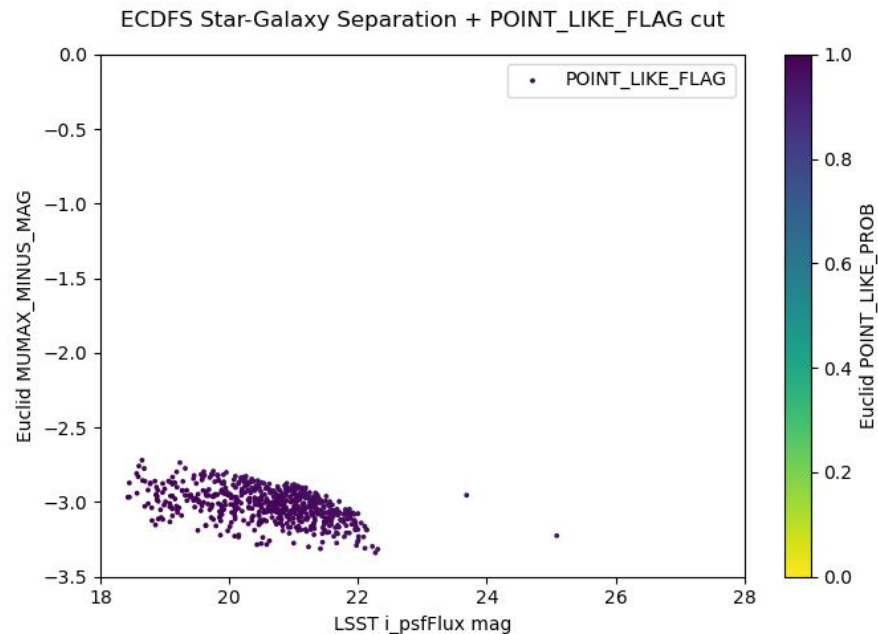
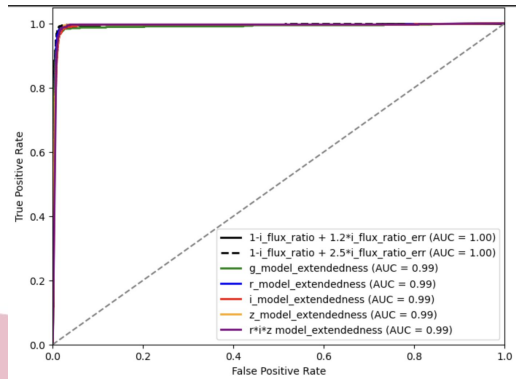
`FWHM <= 1.5`

Using the aperture 2x FWHM for fluxes

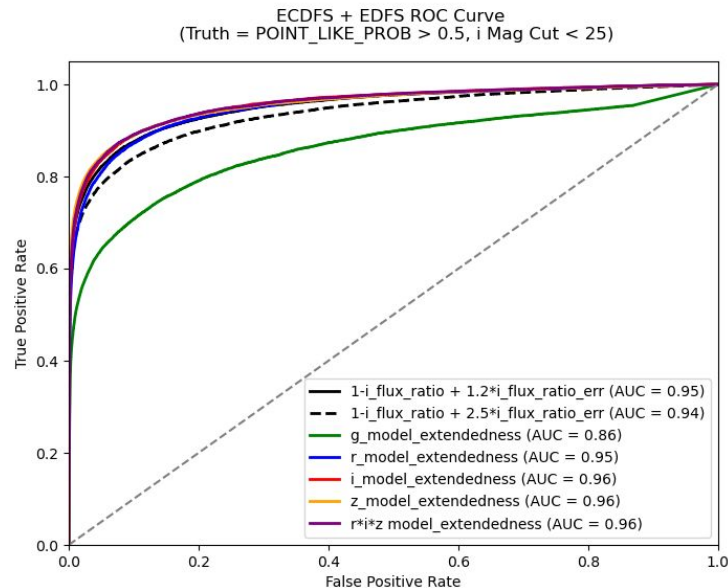
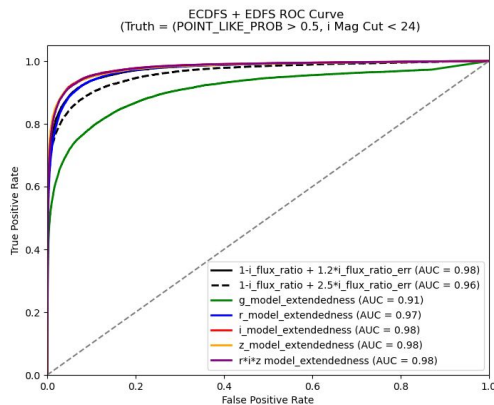
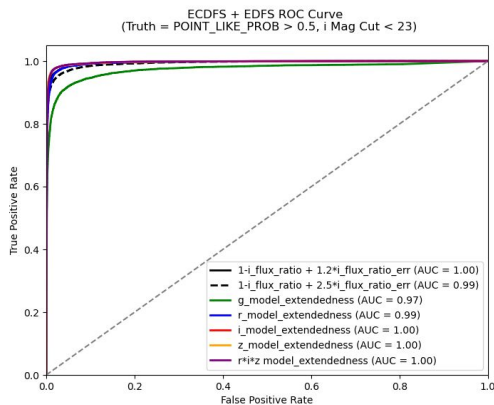
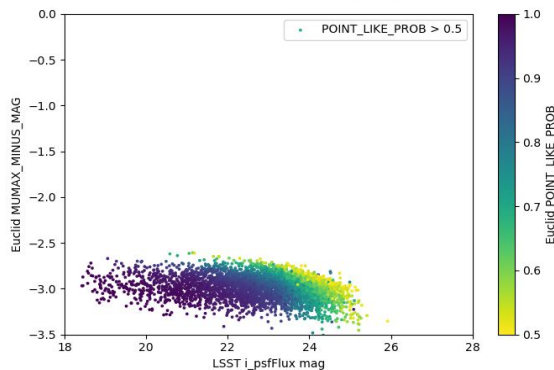
extra slide POINT_LIKE_FLAG

POINT_LIKE_FLAG is the most pure: flag set to 1 for VIS detections with (DET_QUALITY_FLAG==0) AND (POINT_LIKE_PROB>0.96), otherwise set to NaN

It's not complete enough to evaluate our classifiers. The main issue with the Rubin classifiers is that they mix up stars for galaxies so having Euclid call everything a galaxy doesn't help us evaluate them. And the stars that are really easy to identify will be identified by Rubin.



extra slide POINT_LIKE_PROB > 0.5 as truth

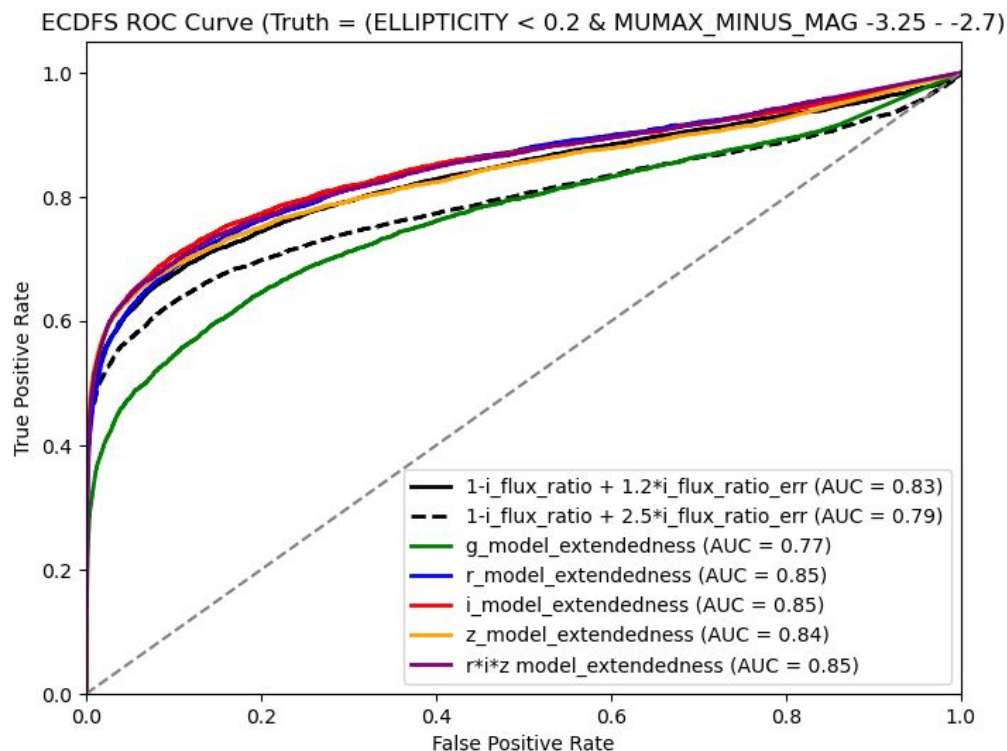


extra slide Zerjal et al's classifier evaluation

However if you use what that Zerjal et al paper suggests, you get this performance

This sample is complete but not as pure, so there are a lot of galaxies our “truth” calls stars

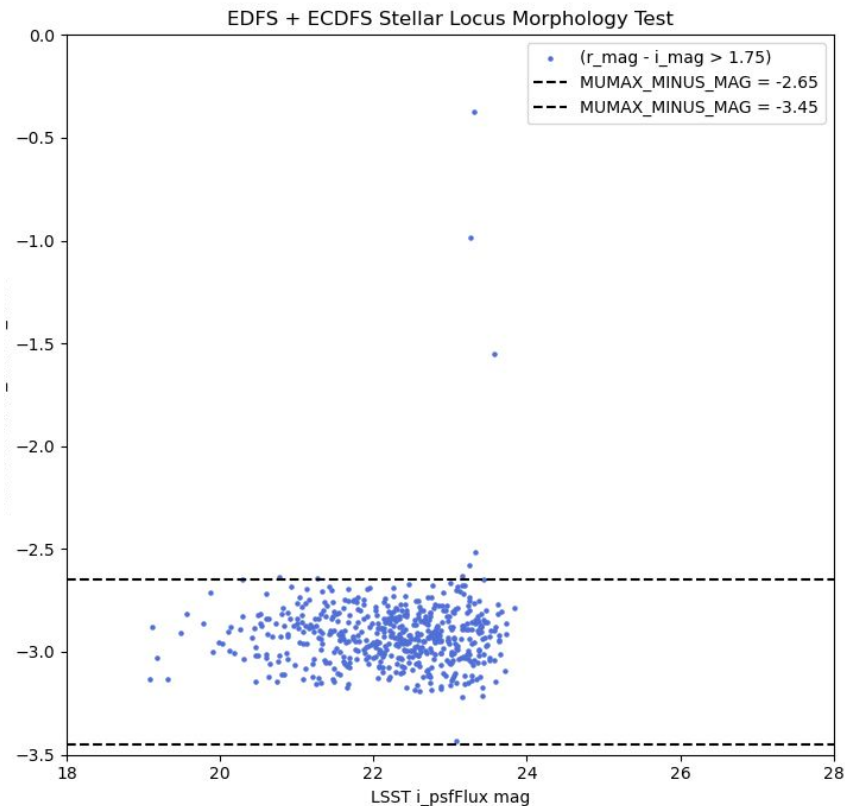
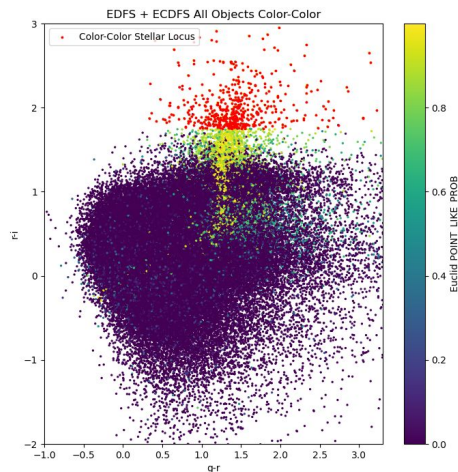
should use a pure sample for truth labels, use complete sample for search



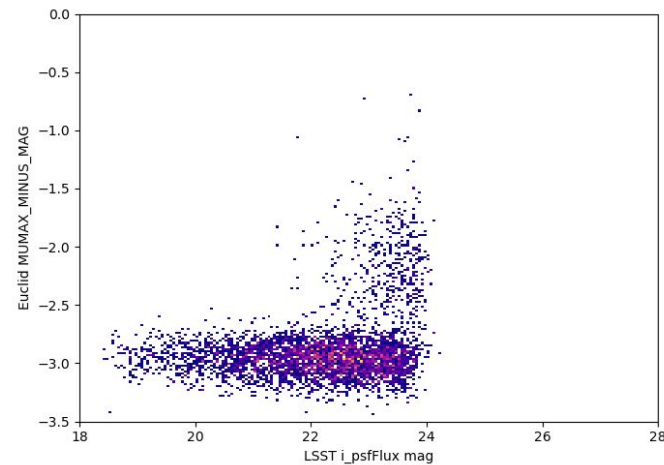
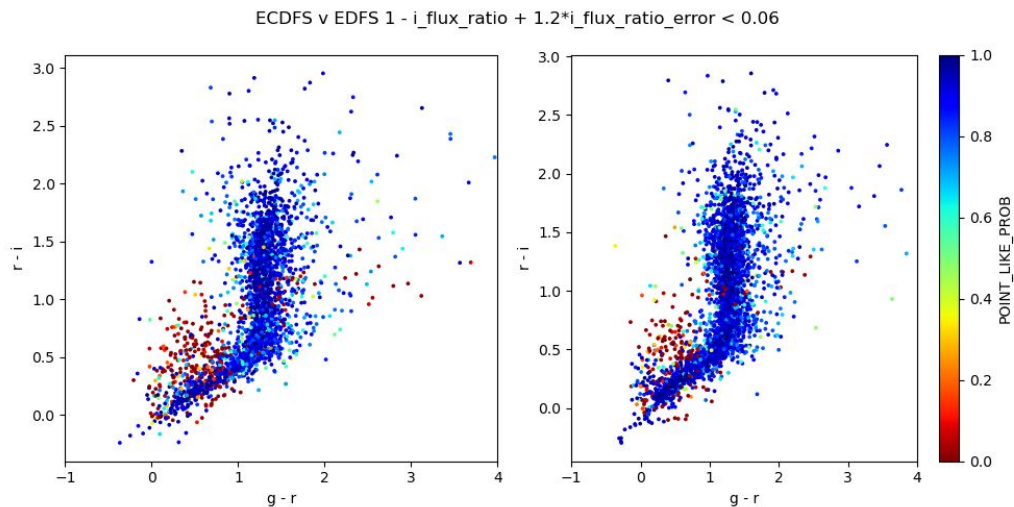
extra slide on MUMAX_MINUS_MAG cut range

Stricter cut in the color-color ->
morphology test

Could justify the bounds
being -2.5 -> -3.5?

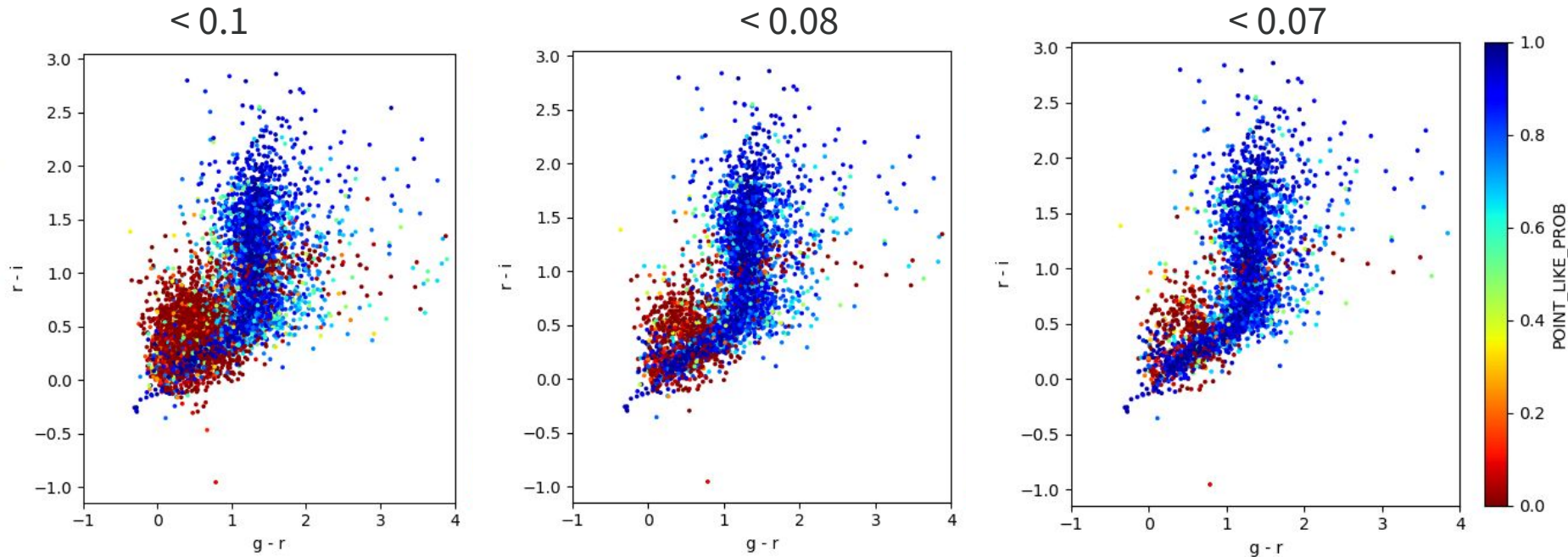


extra slide flux ratio classifier



extra slide flux ratio classifier other thresholds

EDFS: $(1-i_flux_ratio_err) + 1.2*i_flux_ratio_error$



extra slide i_model_extendedness other thresholds

EDFS

